WEB PLATFORM

CSS Layout Visual Guide

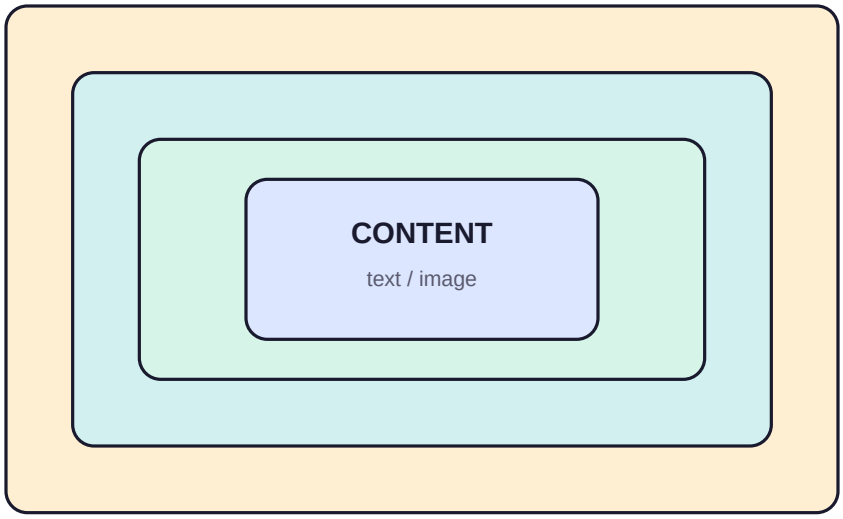
Box model, flexbox, grid, positioning,
and stacking contexts — layout, demystified.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

The Box Model

Every element is nested rectangles: content, padding, border, margin.



box-sizing: border-box -> width includes padding + border. Use it globally.

Margin collapse: adjacent vertical margins merge to the larger one — the classic 'where did my space go'.

STEP 2/6

Positioning Schemes

How an element finds its place — and what it is placed relative to.

static	default flow — top/left ignored
relative	offset from its normal spot; keeps its space
absolute	relative to nearest positioned ancestor; leaves flow
fixed	relative to viewport; stays on scroll
sticky	flows until it hits the threshold, then sticks

STEP 3/6

Flexbox: One-Dimensional Layout

A container distributes space along a main axis; items can grow, shrink, wrap.

flex-direction: row | column

justify-content: main-axis alignment

align-items: cross-axis alignment

flex-wrap: wrap to new lines

FLEX CONTAINER

display:flex — main axis horizontal by default

flex-grow

take a share of free space

flex-shrink

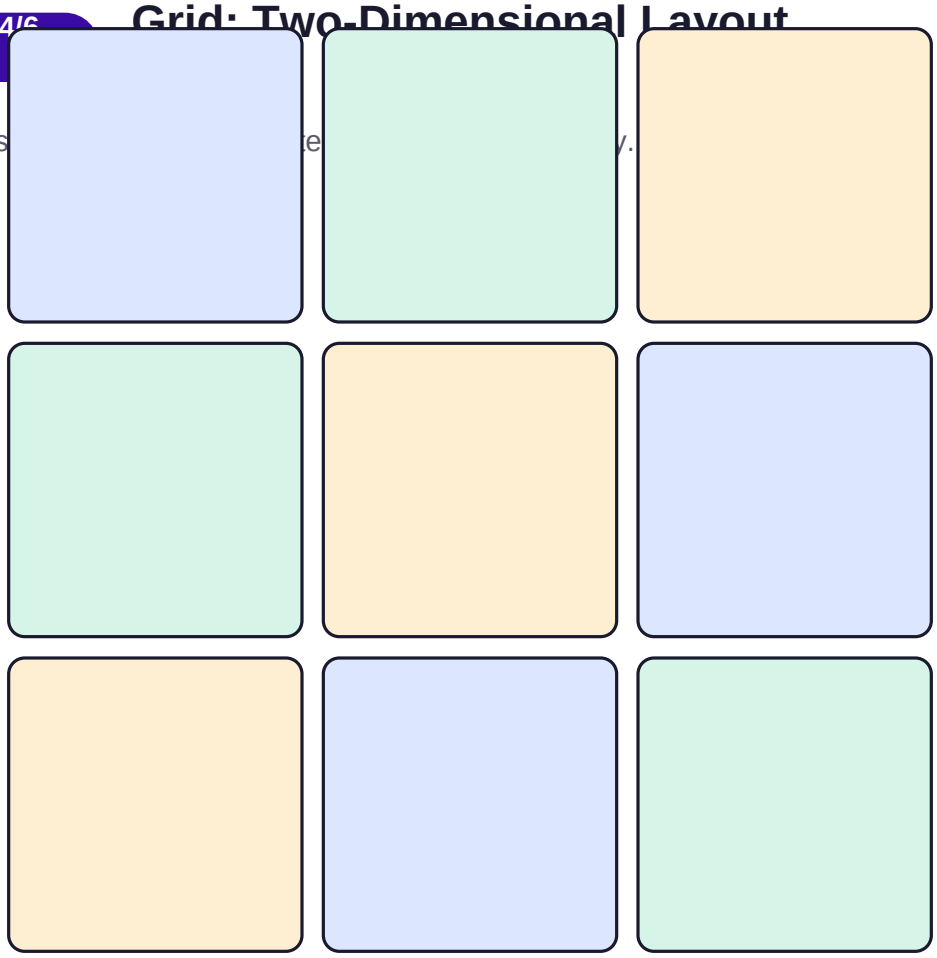
give up space when cramped

Reach for flexbox for navbars, toolbars, centering, and any single row or column.

STEP 4/6

Grid: Two-Dimensional Layout

Define rows



grid-template-columns: 1fr 2fr 1fr — fractional units split free space

grid-template-areas names regions: header header / sidebar main

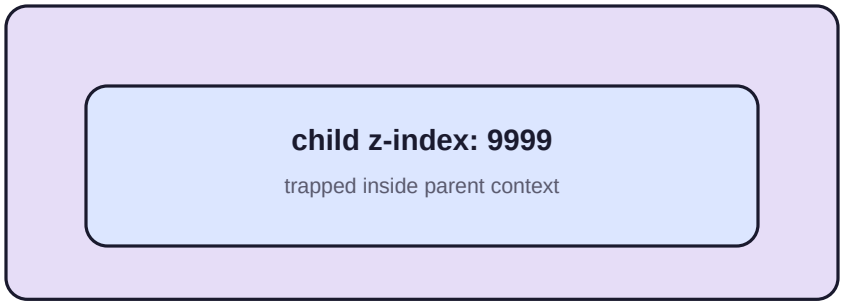
FLEX vs GRID

flex = 1D flow; grid = 2D blueprint. Page skeleton -> grid.

STEP 5/6

Stacking Contexts & z-index

z-index only competes within the same stacking context.



New contexts: positioned + z-index, opacity < 1, transform, filter, will-change.

THE MODAL BUG

'z-index: 99999 not working' = an ancestor created a context

Fix: move the overlay up the DOM or remove the ancestor's context-creating property.

STEP 6/6

Sizing & Responsive Units

Units that adapt instead of breaking.

rem / em	relative to root / parent font size — scales with user settings
vw / vh / %	viewport- and container-relative lengths
fr	grid fractional unit — divides free space
clamp()	clamp(min, preferred, max) — fluid type without breakpoints
container queries	style by the COMPONENT's size, not the viewport

STEP ✓/✓

Key Takeaways

CSS Layout — Remember

- Every element is a box: content, padding, border, margin — use border-box globally.
- Positioning: static flows, relative offsets, absolute/fixed break out, sticky hybridizes.
- Flexbox lays out one dimension (rows or columns); great for components.
- Grid lays out two dimensions; ideal for page skeletons and complex regions.
- z-index only works within a stacking context — check ancestors before raising the number.
- Prefer relative units (rem, %, fr, clamp) and container queries for resilient responsive design.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev